

Programming Fundamentals

Assignment # 3



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Class: BS(Software Engineering)

For Loop

Program no:1

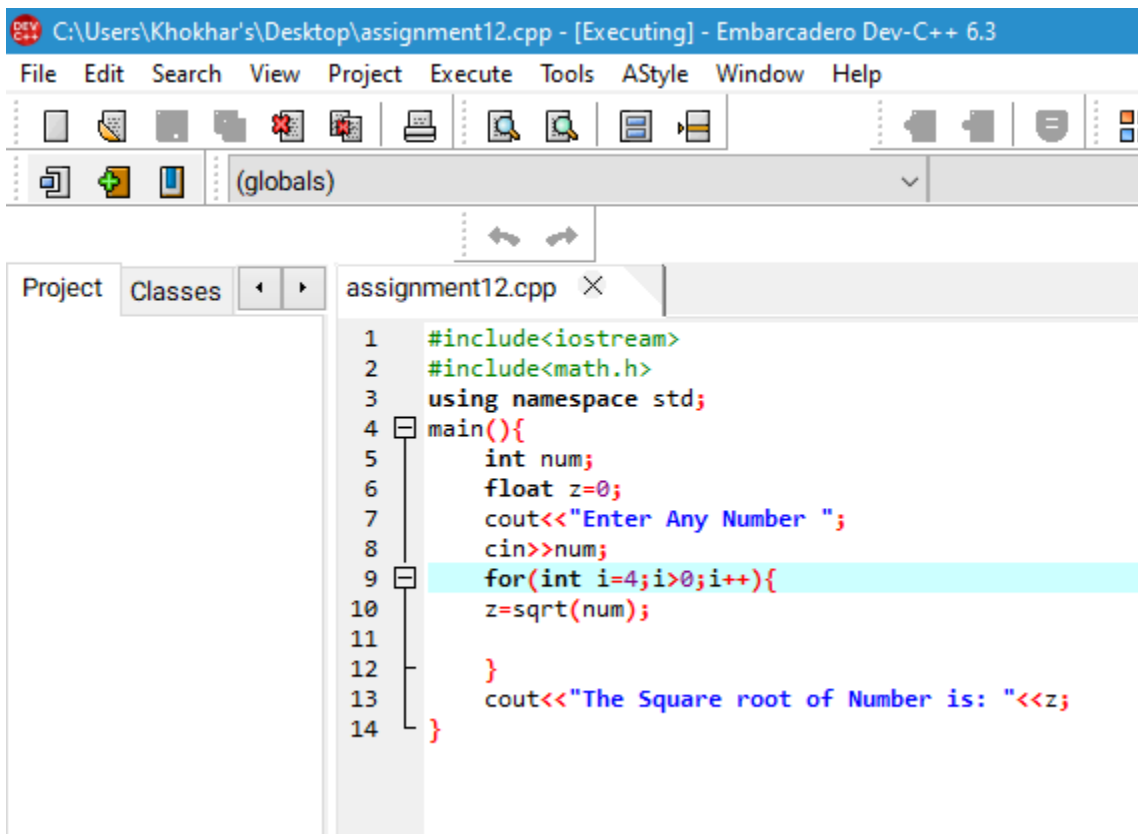
Find Length of a given number?

```
1  #include<iostream>
2  using namespace std;
3  main(){
4  int num,z=0;
5  cout<<"Enter Any Number ";
6  cin>>num;
7  for(;num>0;z++){
8      num=num/10;
9  }
10  cout<<"The Lenght of Number is: "<<z;
11 }
```

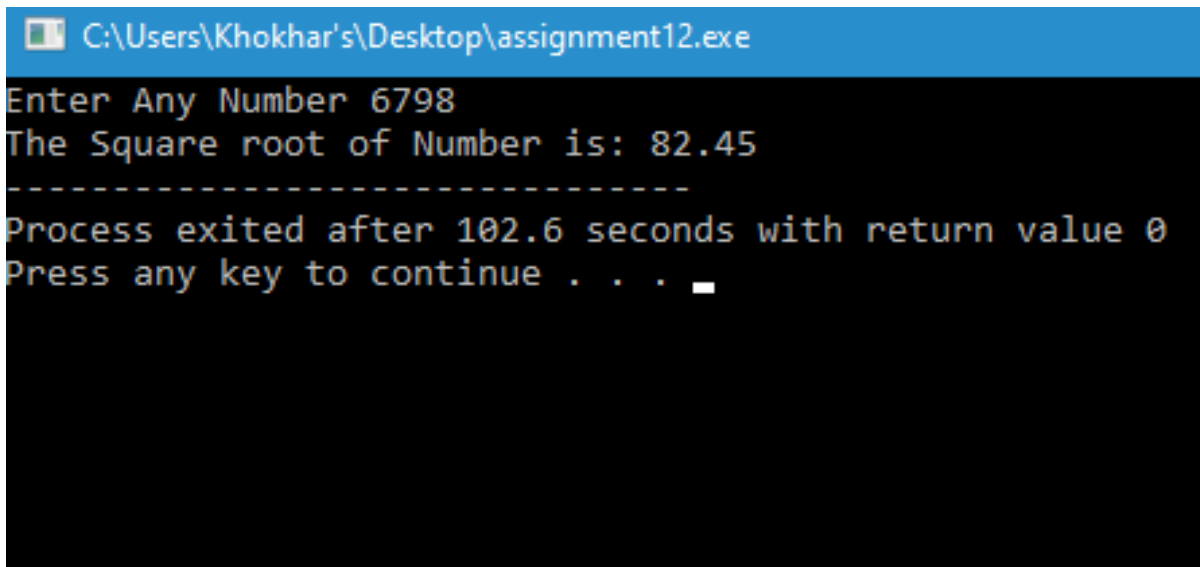
```
C:\Users\Khokhar's\Desktop\assignment12.exe
Enter Any Number 4298
The Lenght of Number is: 4
-----
Process exited after 1.802 seconds with return value 0
Press any key to continue . . .
```

Program no:2

Find the square root of any number?



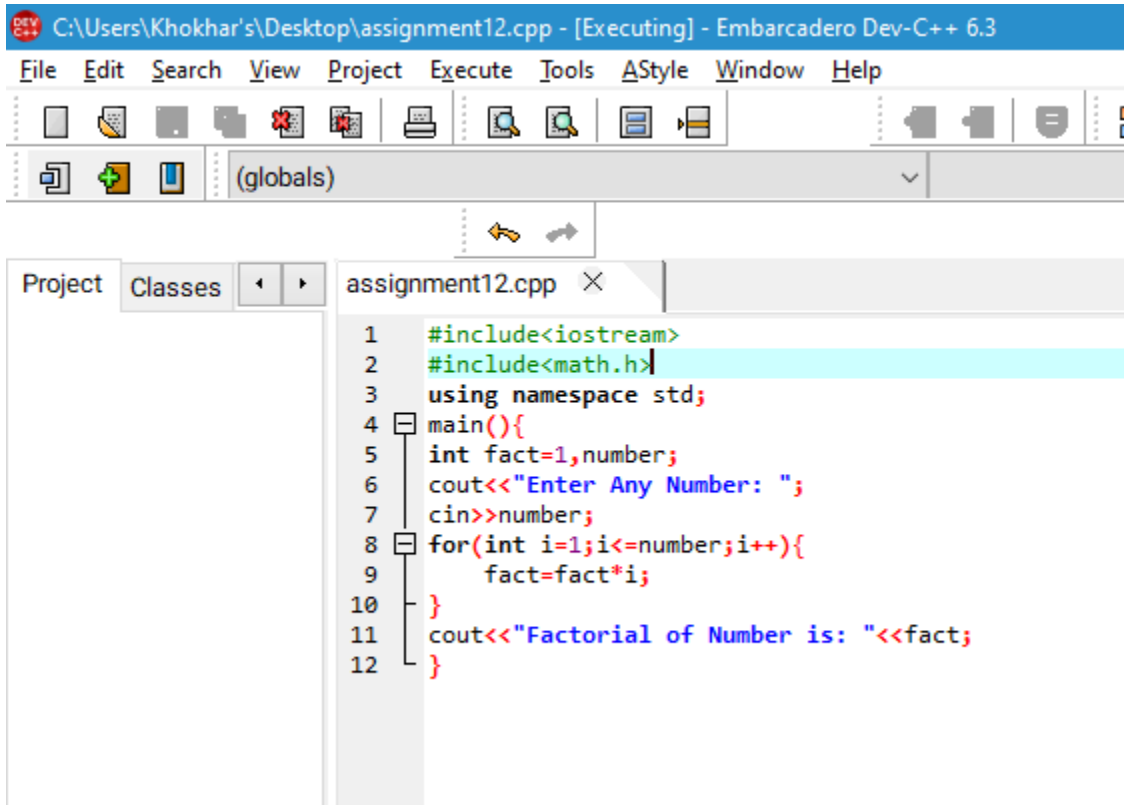
```
1  #include<iostream>
2  #include<math.h>
3  using namespace std;
4  main(){
5      int num;
6      float z=0;
7      cout<<"Enter Any Number ";
8      cin>>num;
9      for(int i=4;i>0;i++){
10         z=sqrt(num);
11     }
12     cout<<"The Square root of Number is: "<<z;
13 }
14 }
```



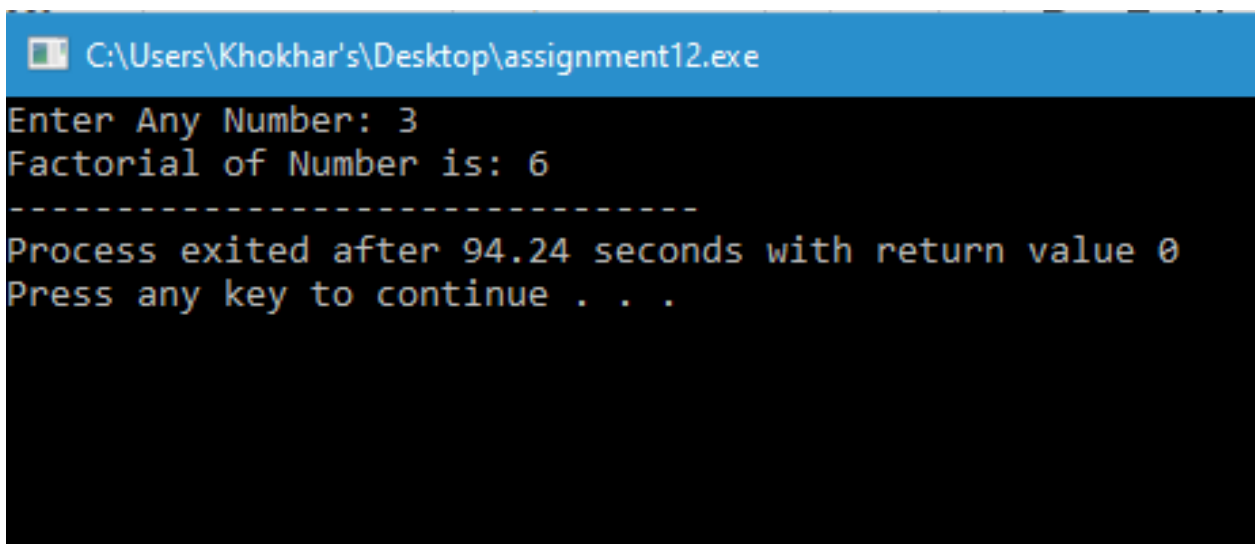
```
C:\Users\Khokhar's\Desktop\assignment12.exe
Enter Any Number 6798
The Square root of Number is: 82.45
-----
Process exited after 102.6 seconds with return value 0
Press any key to continue . . .
```

Program no:3

Find the factorial of any number?



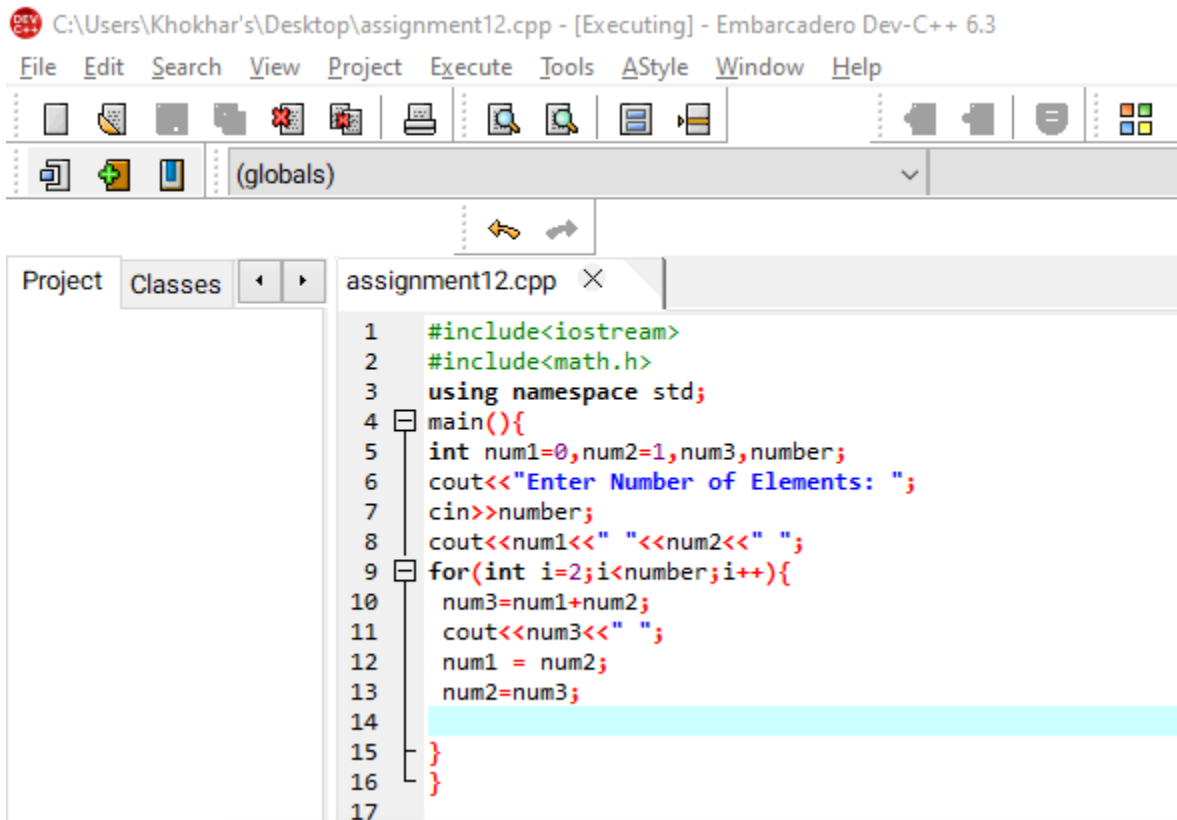
```
C:\Users\Khokhar's\Desktop\assignment12.cpp - [Executing] - Embarcadero Dev-C++ 6.3
File Edit Search View Project Execute Tools AStyle Window Help
(globals)
Project Classes assignment12.cpp
1  #include<iostream>
2  #include<math.h>
3  using namespace std;
4  main(){
5      int fact=1,number;
6      cout<<"Enter Any Number: ";
7      cin>>number;
8      for(int i=1;i<=number;i++){
9          fact=fact*i;
10     }
11     cout<<"Factorial of Number is: "<<fact;
12 }
```



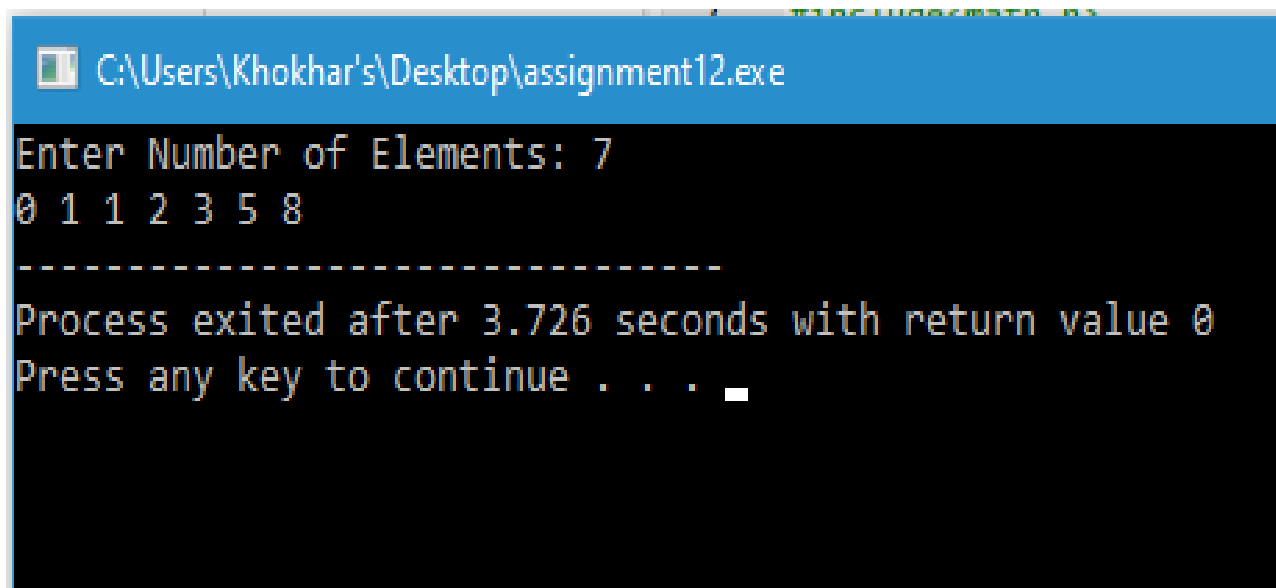
```
C:\Users\Khokhar's\Desktop\assignment12.exe
Enter Any Number: 3
Factorial of Number is: 6
-----
Process exited after 94.24 seconds with return value 0
Press any key to continue . . .
```

Program no:4

Fibonacci Series in C++:



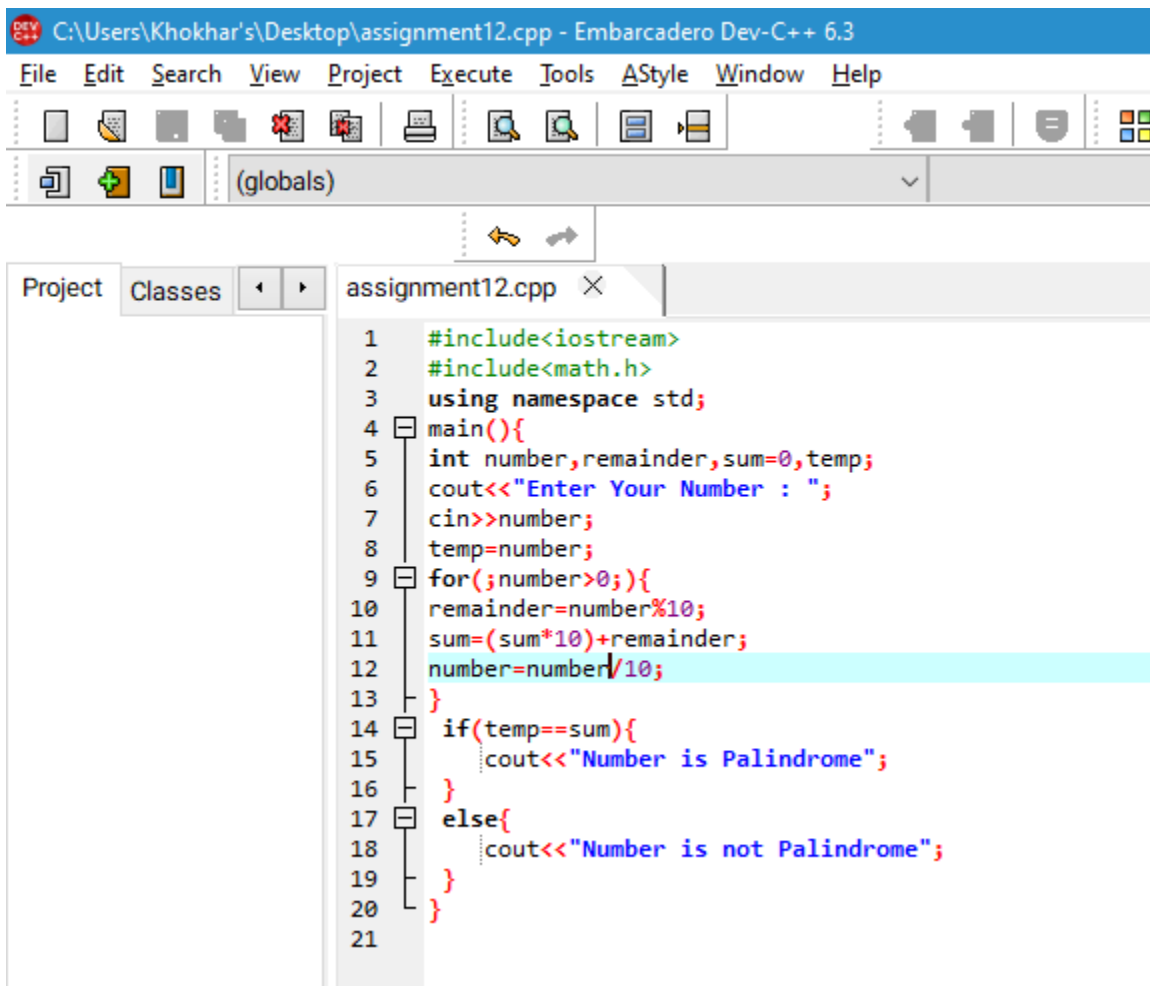
```
1  #include<iostream>
2  #include<math.h>
3  using namespace std;
4  main(){
5      int num1=0,num2=1,num3,number;
6      cout<<"Enter Number of Elements: ";
7      cin>>number;
8      cout<<num1<<" "<<num2<<" ";
9      for(int i=2;i<number;i++){
10         num3=num1+num2;
11         cout<<num3<<" ";
12         num1 = num2;
13         num2=num3;
14
15     }
16 }
17
```



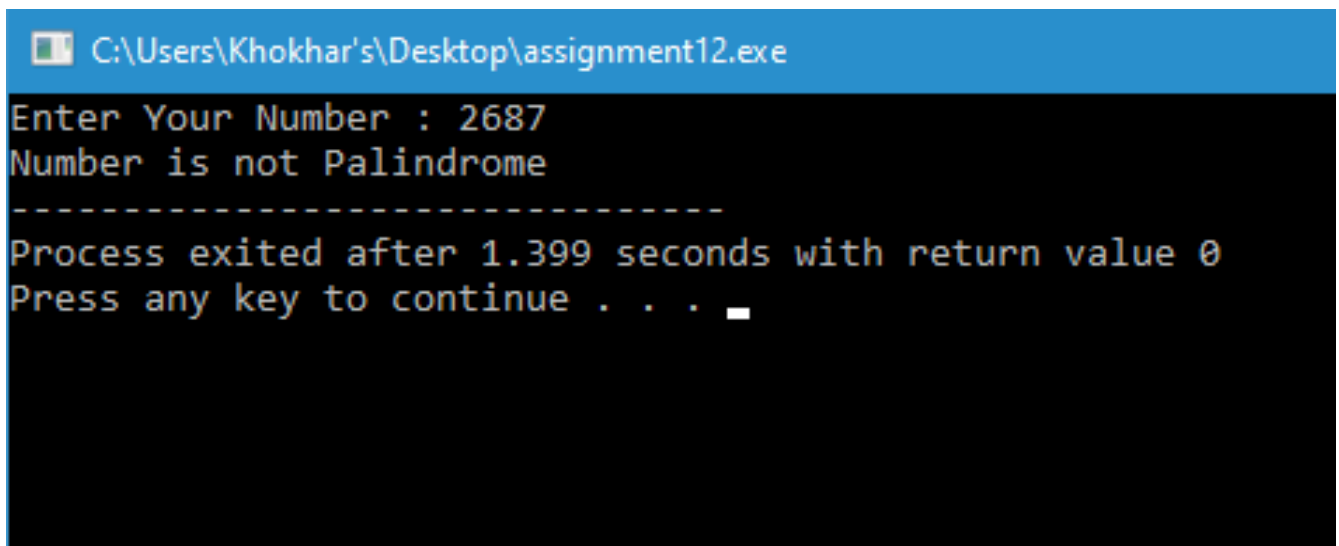
```
C:\Users\Khokhar's\Desktop\assignment12.exe
Enter Number of Elements: 7
0 1 1 2 3 5 8
-----
Process exited after 3.726 seconds with return value 0
Press any key to continue . . .
```

Program no:5

Palindrome Program in C++:



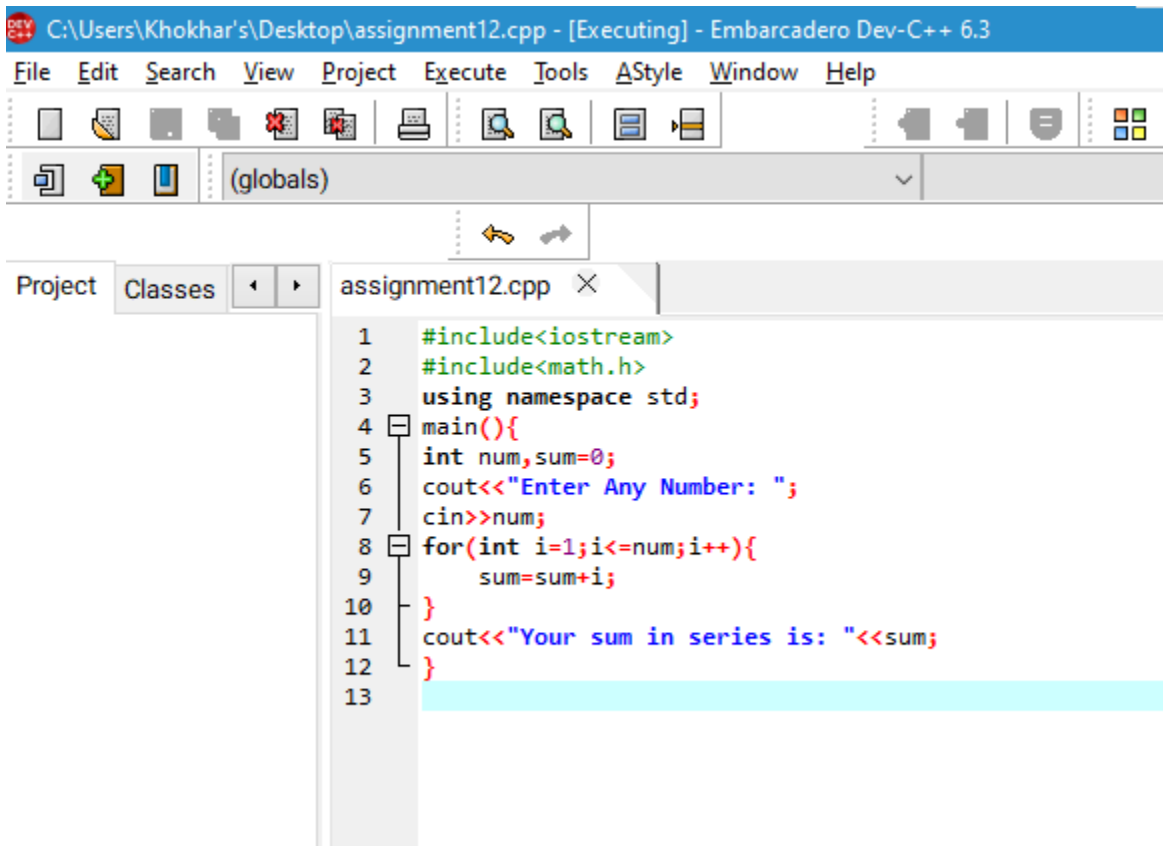
```
1  #include<iostream>
2  #include<math.h>
3  using namespace std;
4  main(){
5      int number,remainder,sum=0,temp;
6      cout<<"Enter Your Number : ";
7      cin>>number;
8      temp=number;
9      for(;number>0;){
10         remainder=number%10;
11         sum=(sum*10)+remainder;
12         number=number/10;
13     }
14     if(temp==sum){
15         cout<<"Number is Palindrome";
16     }
17     else{
18         cout<<"Number is not Palindrome";
19     }
20 }
21
```



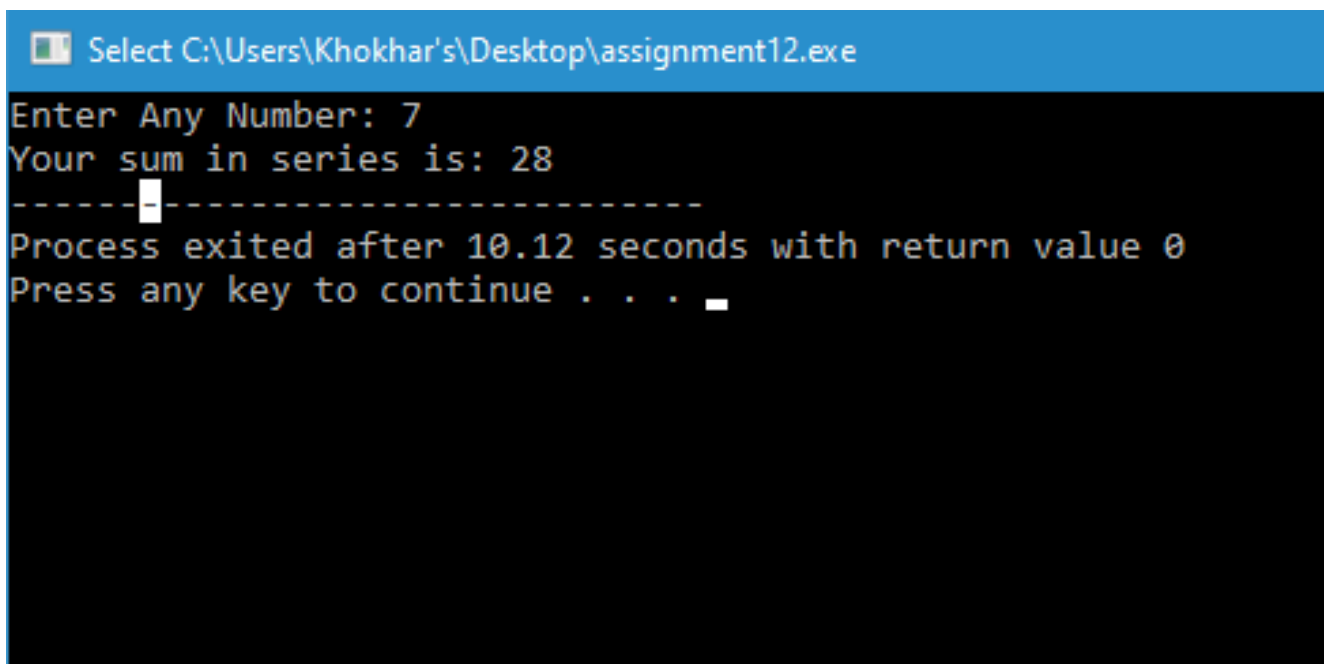
```
C:\Users\Khokhar's\Desktop\assignment12.exe
Enter Your Number : 2687
Number is not Palindrome
-----
Process exited after 1.399 seconds with return value 0
Press any key to continue . . .
```

Program no:6

Write a Program that print sum of the number of series in C++:



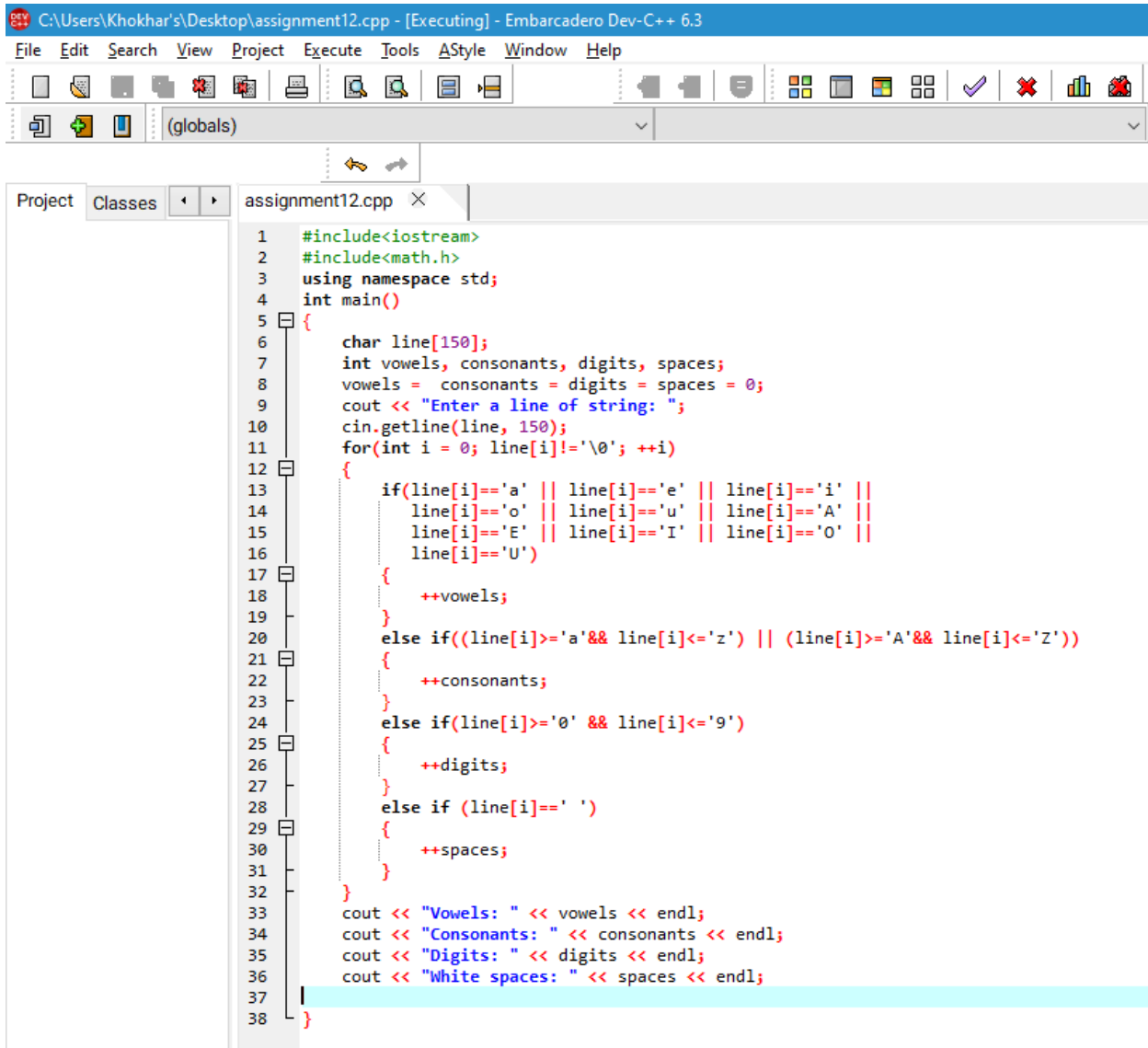
```
C:\Users\Khokhar's\Desktop\assignment12.cpp - [Executing] - Embarcadero Dev-C++ 6.3
File Edit Search View Project Execute Tools AStyle Window Help
(globals)
Project Classes assignment12.cpp
1  #include<iostream>
2  #include<math.h>
3  using namespace std;
4  main(){
5      int num,sum=0;
6      cout<<"Enter Any Number: ";
7      cin>>num;
8      for(int i=1;i<=num;i++){
9          sum=sum+i;
10     }
11     cout<<"Your sum in series is: "<<sum;
12 }
13
```



```
Select C:\Users\Khokhar's\Desktop\assignment12.exe
Enter Any Number: 7
Your sum in series is: 28
-----
Process exited after 10.12 seconds with return value 0
Press any key to continue . . .
```

Program no:7

Write a Program that find the number of vowels, digits, consonants and white spaces in a string:



```
1  #include<iostream>
2  #include<math.h>
3  using namespace std;
4  int main()
5  {
6      char line[150];
7      int vowels, consonants, digits, spaces;
8      vowels = consonants = digits = spaces = 0;
9      cout << "Enter a line of string: ";
10     cin.getline(line, 150);
11     for(int i = 0; line[i]!='\0'; ++i)
12     {
13         if(line[i]=='a' || line[i]=='e' || line[i]=='i' ||
14            line[i]=='o' || line[i]=='u' || line[i]=='A' ||
15            line[i]=='E' || line[i]=='I' || line[i]=='O' ||
16            line[i]=='U')
17         {
18             ++vowels;
19         }
20         else if((line[i]>='a' && line[i]<='z') || (line[i]>='A' && line[i]<='Z'))
21         {
22             ++consonants;
23         }
24         else if(line[i]>='0' && line[i]<='9')
25         {
26             ++digits;
27         }
28         else if (line[i]==' ')
29         {
30             ++spaces;
31         }
32     }
33     cout << "Vowels: " << vowels << endl;
34     cout << "Consonants: " << consonants << endl;
35     cout << "Digits: " << digits << endl;
36     cout << "White spaces: " << spaces << endl;
37
38 }
```

```
C:\Users\Khokhar's\Desktop\assignment12.exe

Enter a line of string: Hi! are you ready.
Vowels: 7
Consonants: 6
Digits: 0
White spaces: 3

-----
Process exited after 26.87 seconds with return value 0
Press any key to continue . . .
```

Program no:8

Write a Program that display reverse of a number:

```
C:\Users\Khokhar's\Desktop\assignment12.cpp - [Executing] - Embarcadero Dev-C++ 6.3

File Edit Search View Project Execute Tools AStyle Window Help

(globals)

Project Classes assignment12.cpp X

1  #include<iostream>
2  #include<math.h>
3  using namespace std;
4  int main() {
5      int n, rev_num = 0, remainder;
6
7      cout << "Enter an integer: ";
8      cin >> n;
9      for(;n>0;) {
10         remainder = n % 10;
11         rev_num = rev_num * 10 + remainder;
12         n /= 10;
13     }
14
15     cout << "The Number is Reversed Now = " << rev_num;
16 }
17
```



```
C:\Users\Khokhar's\Desktop\assignment12.exe
Enter an integer: 3484
The Number is Reversed Now = 4843
-----
Process exited after 2.817 seconds with return value 0
Press any key to continue . . .
```

Program no:9

Write a Program that lets the user enter the total rainfall for each of 12 months:

```
C:\Users\Khokhar's\Desktop\assignment12.cpp - Embarcadero Dev-C++ 6.3
File Edit Search View Project Execute Tools AStyle Window Help
(globals) TDM-GCC 9.2.0 32-bit Release

assignment12.cpp
1  #include<iostream>
2  #include<math.h>
3  #include <string>
4  #include <iomanip>
5  using namespace std;
6  int main ()
7  {
8      const int ALLMONTHS = 12;
9      string name[ALLMONTHS]= {"January", "February", "March", "April", "May", "June", "July", "August", "September", "October", "November", "December"};
10     int LoopVariable= 0;
11     double rain[ALLMONTHS];
12     double avg;
13     double year=0;
14     double maximum;
15     string highMonth;
16     double minimum;
17     string lowMonth;
18
19     for(LoopVariable = 0; LoopVariable < ALLMONTHS; LoopVariable++) // ask user to enter amount of rainfall for each month
20     {
21         cout << "How many inches of rain does " << name[LoopVariable];
22         cout << " have? \n";
23         cin >> rain[LoopVariable];
24         while (rain[LoopVariable] < 0)
25         {
26             cout << "Please enter a number greater than 0." << endl;
27             cin >> rain[LoopVariable];
28         }
29     }
30     for(int LoopVariable=0; LoopVariable<ALLMONTHS; LoopVariable++) // totals up all the rainfall
31     year += rain[LoopVariable];
32     avg = year / ALLMONTHS;
33
34     for( int LoopVariable = 0; LoopVariable < ALLMONTHS; LoopVariable++) //spits out each month with its amount of rain
35     {
36         cout << name[LoopVariable];
37         cout << " has ";
38         cout << rain[LoopVariable] << " inches of rainfall.\n";
39     }
```

C:\Users\Khokhar's\Desktop\assignment12.cpp - Embarcadero Dev-C++ 6.3

File Edit Search View Project Execute Tools AStyle Window Help

(globals) TDM-GCC 9.2.0 32-bit

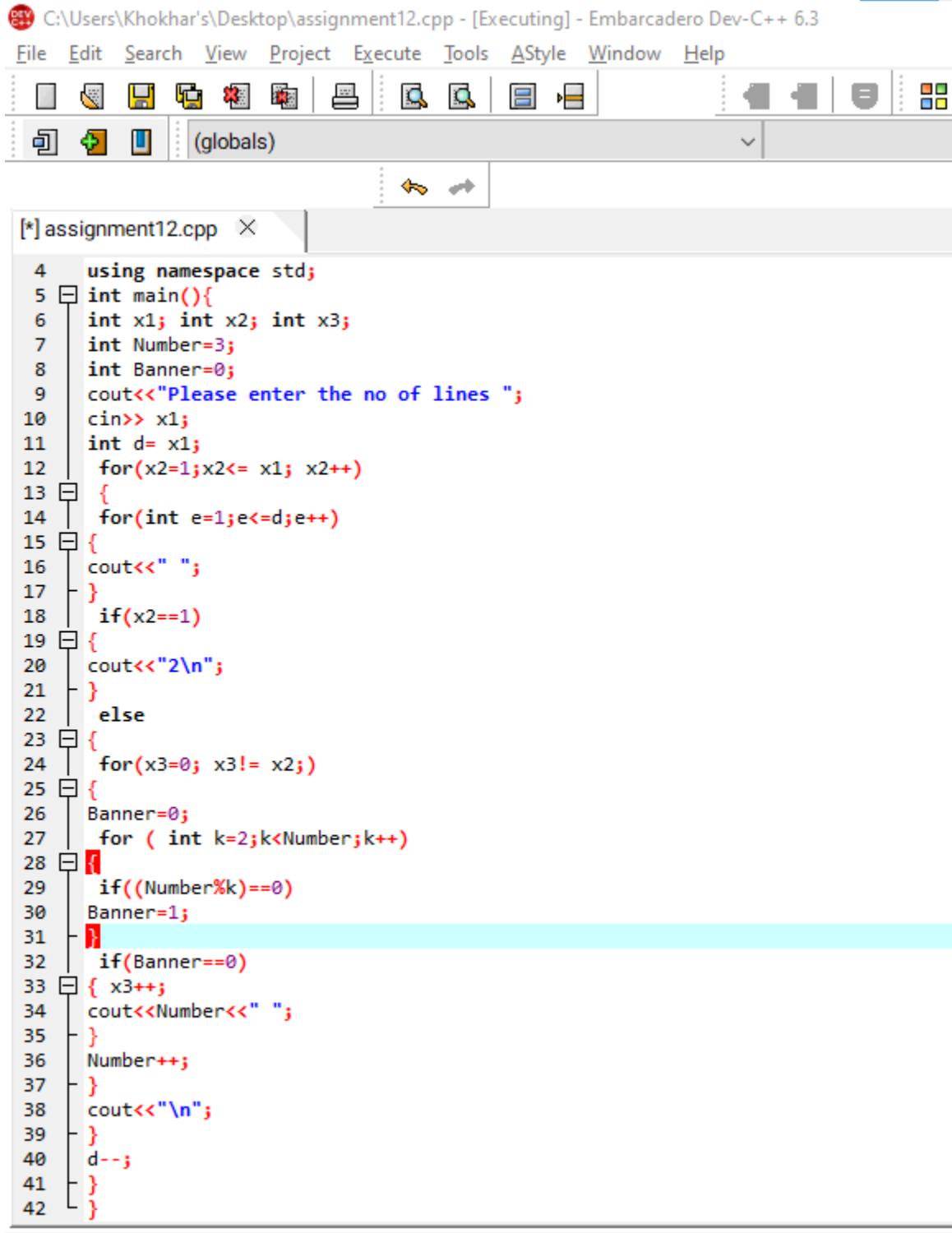
assignment12.cpp

```
37 cout<< " has ";
38 cout<< rain[LoopVariable] << " inches of rainfall.\n";
39 }
40
41 maximum = rain[0]; // finds month with the maximum amount of rain
42
43 for (LoopVariable = 1 ;LoopVariable < ALLMONTHS; LoopVariable++)
44 {
45     if (rain[LoopVariable] > maximum)
46     {
47         highMonth = name[LoopVariable];
48         maximum = rain[LoopVariable];
49     }
50 }
51
52 minimum = rain[0]; // finds month with the least amount of rain
53
54 for (LoopVariable = 1 ;LoopVariable < ALLMONTHS; LoopVariable++)
55 {
56     if (rain[LoopVariable] < minimum)
57     {
58         lowMonth = name[LoopVariable];
59         minimum = rain[LoopVariable];
60     }
61 }
62
63 cout << endl;
64
65 cout << setprecision(2) << fixed;
66
67 cout <<"Total Rainfall in the year is total" <<year << " inches" << endl;
68
69 cout <<"The average monthly rainfall is total" << avg << " inches"<< endl;
70
71 cout <<"The month with the maximum amount of rainfall is total " << highMonth << " with " << maximum<< " inches."<< endl;
72
73 cout <<"The month with the minimum amount of rainfall is total" << lowMonth << " with " << minimum<< " inches."<< endl;
74
75
```

```
C:\Users\Khokhar's\Desktop\assignment12.exe
File
How many inches of rain does January have?
643
How many inches of rain does February have?
435
How many inches of rain does March have?
24
How many inches of rain does April have?
54
37 How many inches of rain does May have?
38 234
39 How many inches of rain does June have?
40 43
41 How many inches of rain does July have?
42 23
43 How many inches of rain does August have?
44 435
45 How many inches of rain does September have?
46 6
47 How many inches of rain does October have?
48 3
49 How many inches of rain does November have?
50 565
51 How many inches of rain does December have?
52 345
53 January has 643 inches of rainfall.
54 February has 435 inches of rainfall.
55 March has 24 inches of rainfall.
56 April has 54 inches of rainfall.
57 May has 234 inches of rainfall.
58 June has 43 inches of rainfall.
59 July has 23 inches of rainfall.
60 August has 435 inches of rainfall.
61 September has 6 inches of rainfall.
62 October has 3 inches of rainfall.
63 November has 565 inches of rainfall.
64 December has 345 inches of rainfall.
65
66 Total Rainfall in the year is total2810.00 inches
67 The average monthly rainfall is total234.17 inches
68 The month with the maximum amount of rainfall is total with 643.00 inches.
69 The month with the minimum amount of rainfall is totalOctober with 3.00 inches.
70
71 -----
72 Process exited after 10.31 seconds with return value 0
73 Press any key to continue . . .
```

Program no:10

Write a Program that print a triangle of prime numbers up to given number of lines of the triangle:



```
C:\Users\Khokhar's\Desktop\assignment12.cpp - [Executing] - Embarcadero Dev-C++ 6.3
File Edit Search View Project Execute Tools AStyle Window Help
(globals)
[*] assignment12.cpp
4 using namespace std;
5 int main(){
6 int x1; int x2; int x3;
7 int Number=3;
8 int Banner=0;
9 cout<<"Please enter the no of lines ";
10 cin>> x1;
11 int d= x1;
12 for(x2=1;x2<= x1; x2++)
13 {
14 for(int e=1;e<=d;e++)
15 {
16 cout<<" ";
17 }
18 if(x2==1)
19 {
20 cout<<"2\n";
21 }
22 else
23 {
24 for(x3=0; x3!= x2;)
25 {
26 Banner=0;
27 for ( int k=2;k<Number;k++)
28 {
29 if((Number%k)==0)
30 Banner=1;
31 }
32 if(Banner==0)
33 { x3++;
34 cout<<Number<<" ";
35 }
36 Number++;
37 }
38 cout<<"\n";
39 }
40 d--;
41 }
42 }
```

```
C:\Users\Khokhar's\Desktop\assignment12.exe

Please enter the no of lines 4
2
3 5
7 11 13
17 19 23 29

-----
Process exited after 4.004 seconds with return value 0
Press any key to continue . . .
```

Program no:11

Write a Program that print all numbers between a and b (a and b inclusive) using for loop:

```
C:\Users\Khokhar's\Desktop\assignment12.cpp - [Executing] - Embarcadero Dev-C++ 6.3
File Edit Search View Project Execute Tools AStyle Window Help
(globals)
assignment12.cpp
1  #include<iostream>
2  using namespace std;
3  int main()
4  {
5      int a=2;
6      int b=100;
7      for(int loop=a; loop<=b; loop++)
8      {
9          cout << "All numbers between a and b are:"<<loop<<endl;
10     }
11
12 }
```

```
C:\Users\Khokhar's\Desktop\assignment12.exe
All numbers between a and b are:76
All numbers between a and b are:77
All numbers between a and b are:78
All numbers between a and b are:79
All numbers between a and b are:80
All numbers between a and b are:81
All numbers between a and b are:82
All numbers between a and b are:83
All numbers between a and b are:84
All numbers between a and b are:85
All numbers between a and b are:86
All numbers between a and b are:87
All numbers between a and b are:88
All numbers between a and b are:89
All numbers between a and b are:90
All numbers between a and b are:91
All numbers between a and b are:92
All numbers between a and b are:93
All numbers between a and b are:94
All numbers between a and b are:95
All numbers between a and b are:96
All numbers between a and b are:97
All numbers between a and b are:98
All numbers between a and b are:99
All numbers between a and b are:100

-----
Process exited after 0.235 seconds with return value 0
Press any key to continue . . .
```

Program no:12

Write a simple C++ Program to find mean value:

DEV C++ C:\Users\Khokhar's\Desktop\assignment12.cpp - [Executing] - Embarcadero Dev-C++ 6.3

File Edit Search View Project Execute Tools AStyle Window Help

(globals)

assignment12.cpp

```
1  #include<iostream>
2  using namespace std;
3  int main() {
4      int num_count;
5      double sum, mean;
6      cout<<"Enter the number of values: ";
7      cin>>num_count;
8      for(int i=0; i<num_count; i++) {
9          double num;
10         cout<<"Enter value "<<i+1<<": ";
11         cin>>num;
12         sum+=num;
13     }
14     mean=sum/num_count;
15     cout<<"The mean value is: "<<mean<<endl;
16 }
```

C:\Users\Khokhar's\Desktop\assignment12.exe

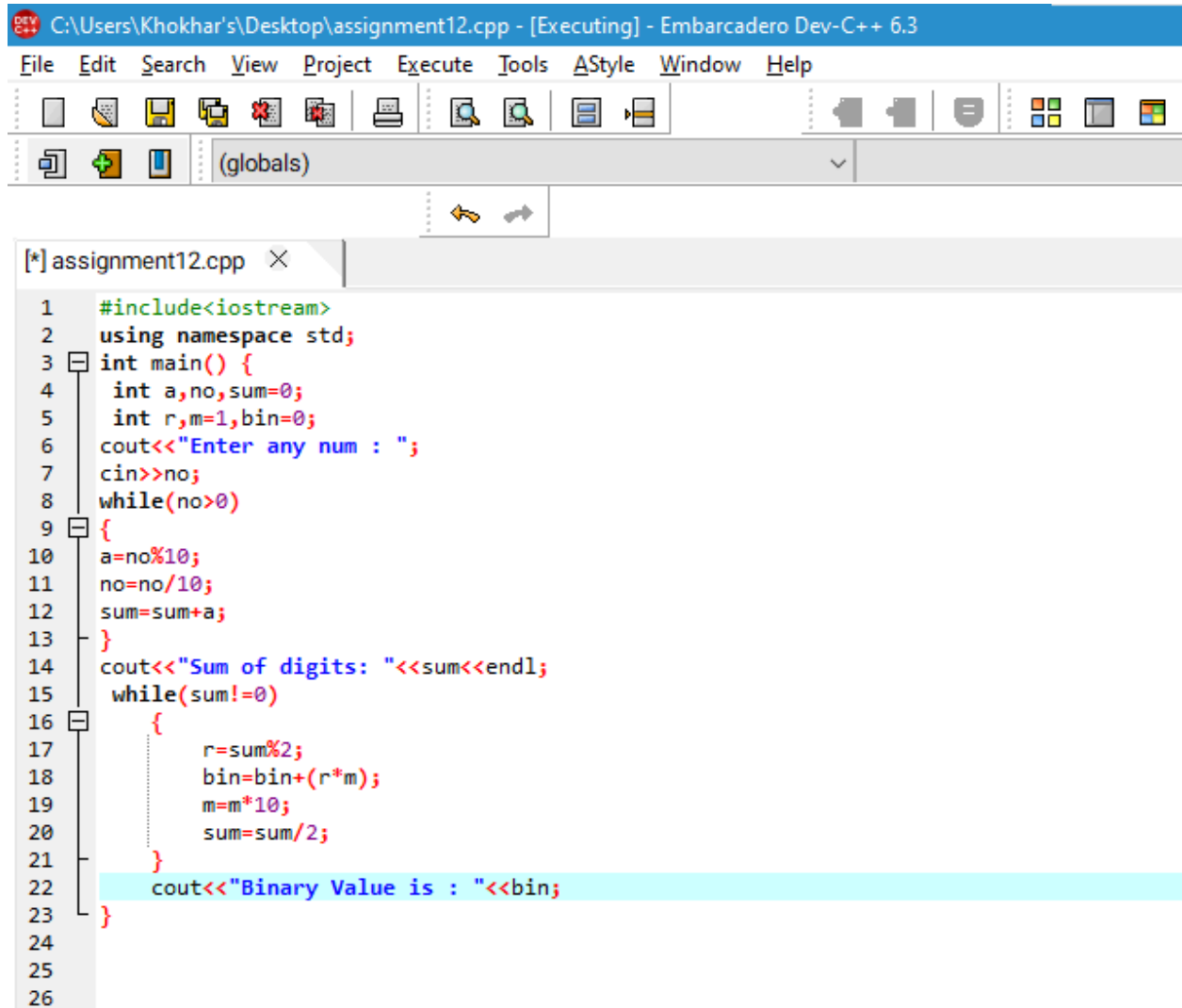
```
Enter the number of values: 5
Enter value 1: 23
Enter value 2: 45
Enter value 3: 23
Enter value 4: 54
Enter value 5: 65
The mean value is: 42

-----
Process exited after 16.3 seconds with return value 0
Press any key to continue . . .
```

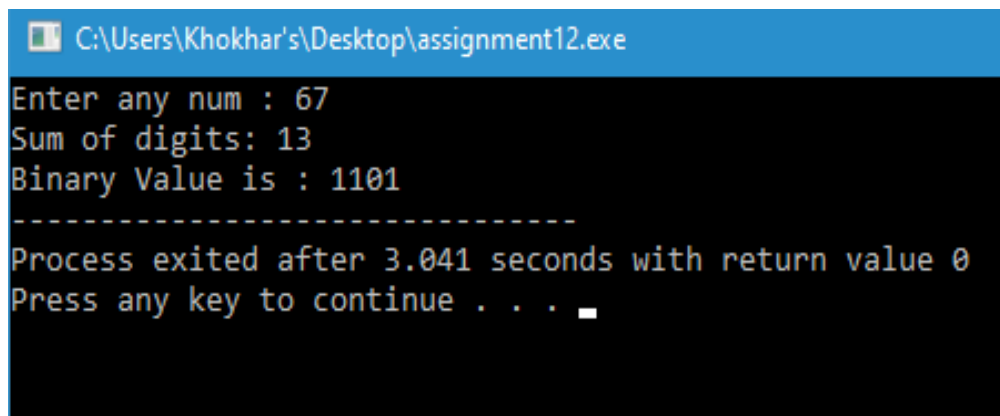
While Loop

Program no:1

Program of sum of all digits of a number that converts a number into binary in C++.



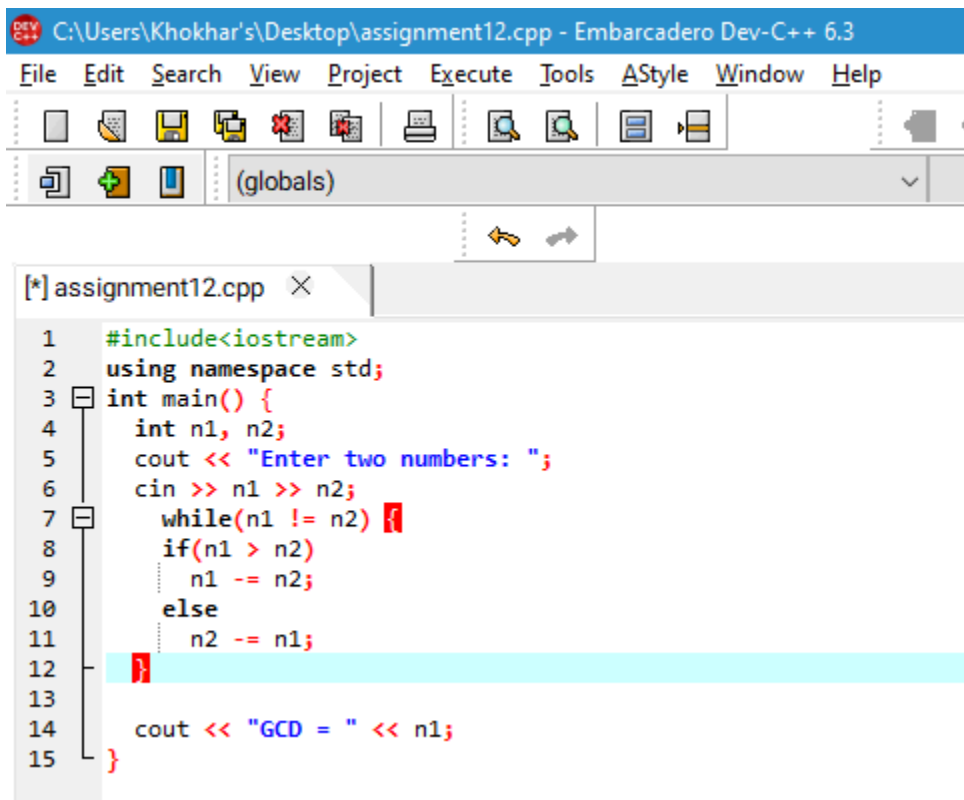
```
1  #include<iostream>
2  using namespace std;
3  int main() {
4      int a,no,sum=0;
5      int r,m=1,bin=0;
6      cout<<"Enter any num : ";
7      cin>>no;
8      while(no>0)
9      {
10         a=no%10;
11         no=no/10;
12         sum=sum+a;
13     }
14     cout<<"Sum of digits: "<<sum<<endl;
15     while(sum!=0)
16     {
17         r=sum%2;
18         bin=bin+(r*m);
19         m=m*10;
20         sum=sum/2;
21     }
22     cout<<"Binary Value is : "<<bin;
23 }
24
25
26
```



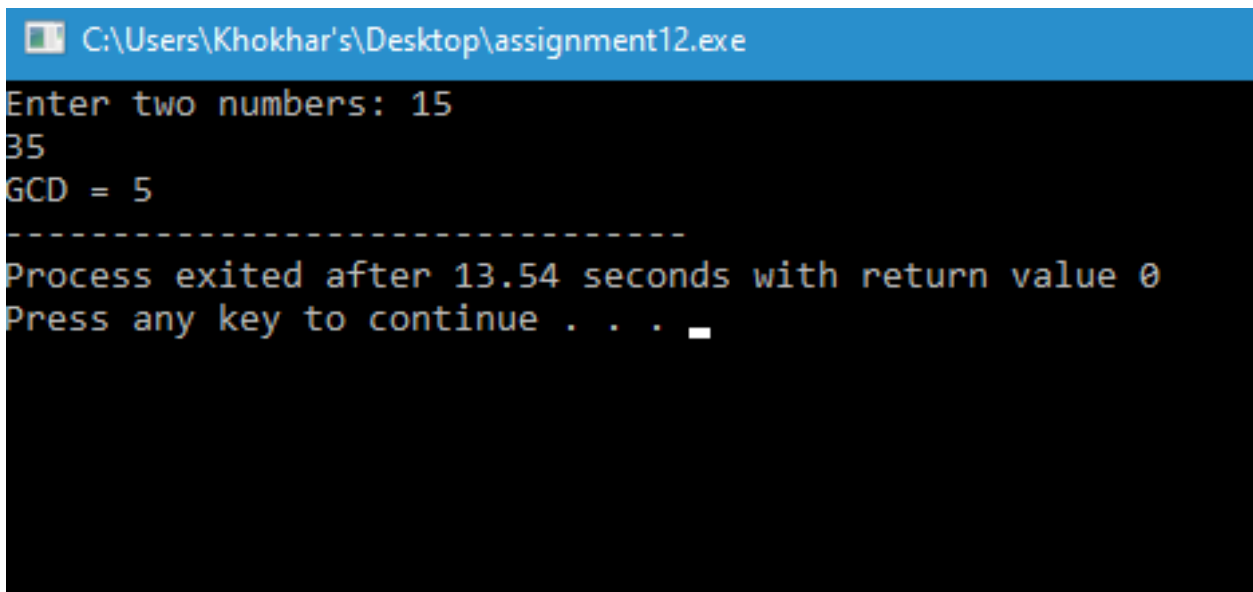
```
C:\Users\Khokhar's\Desktop\assignment12.exe
Enter any num : 67
Sum of digits: 13
Binary Value is : 1101
-----
Process exited after 3.041 seconds with return value 0
Press any key to continue . . .
```


Program no:2

Program in C++ with explanation to find the greatest common Division(GCD):



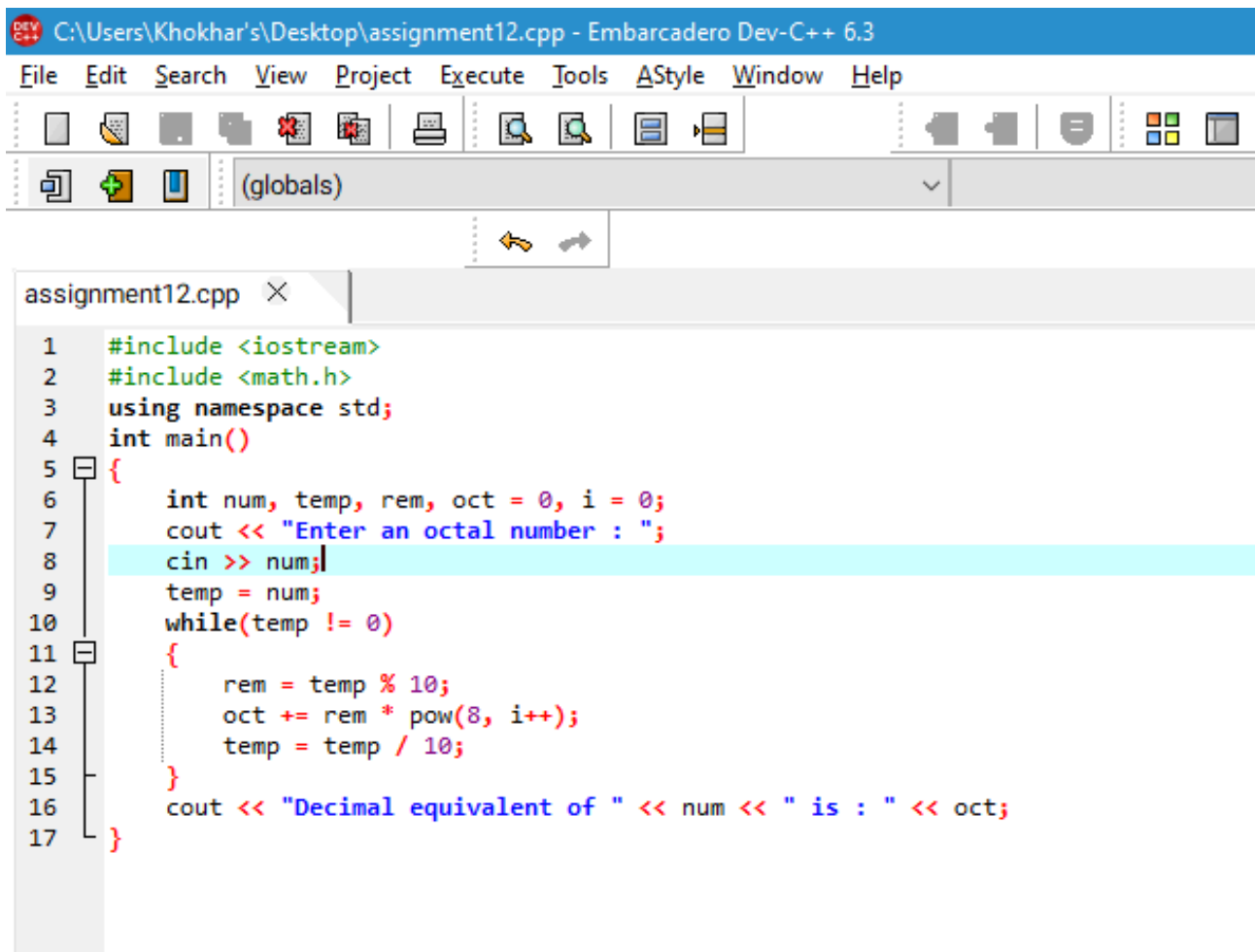
```
1  #include<iostream>
2  using namespace std;
3  int main() {
4      int n1, n2;
5      cout << "Enter two numbers: ";
6      cin >> n1 >> n2;
7      while(n1 != n2) {
8          if(n1 > n2)
9              n1 -= n2;
10         else
11             n2 -= n1;
12     }
13
14     cout << "GCD = " << n1;
15 }
```



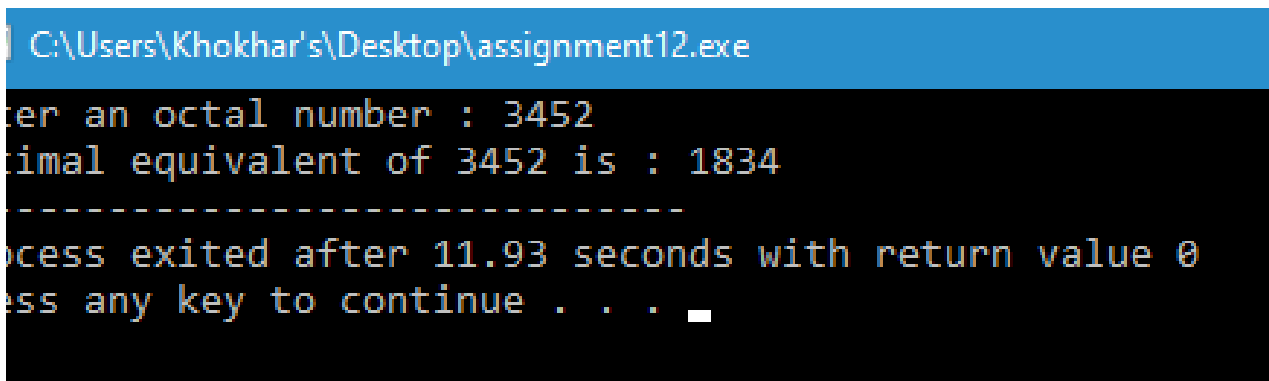
```
C:\Users\Khokhar's\Desktop\assignment12.exe
Enter two numbers: 15
35
GCD = 5
-----
Process exited after 13.54 seconds with return value 0
Press any key to continue . . .
```

Program no:3

Program in C++ that converts Octal to decimal number using while loop:



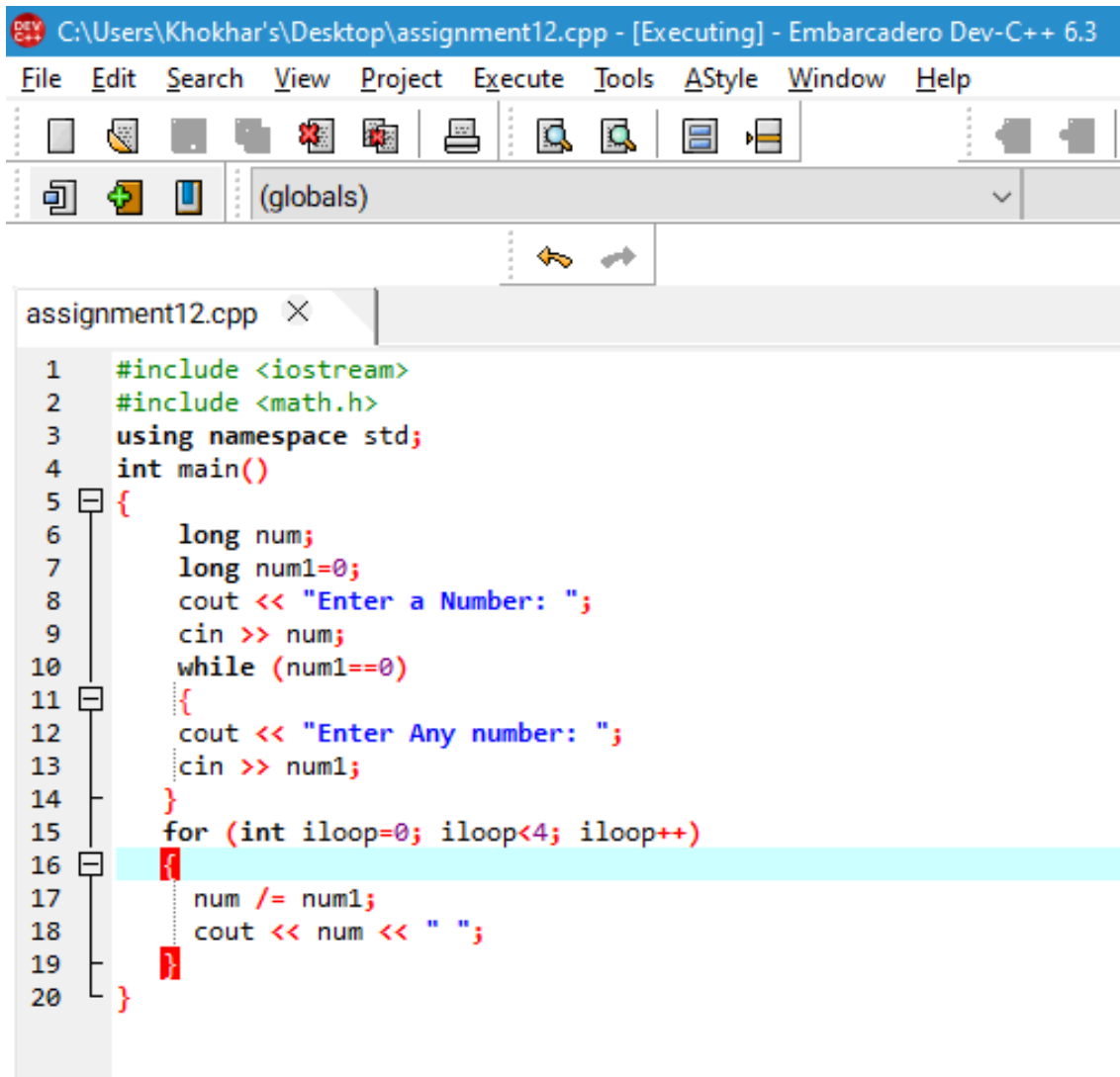
```
1  #include <iostream>
2  #include <math.h>
3  using namespace std;
4  int main()
5  {
6      int num, temp, rem, oct = 0, i = 0;
7      cout << "Enter an octal number : ";
8      cin >> num;
9      temp = num;
10     while(temp != 0)
11     {
12         rem = temp % 10;
13         oct += rem * pow(8, i++);
14         temp = temp / 10;
15     }
16     cout << "Decimal equivalent of " << num << " is : " << oct;
17 }
```



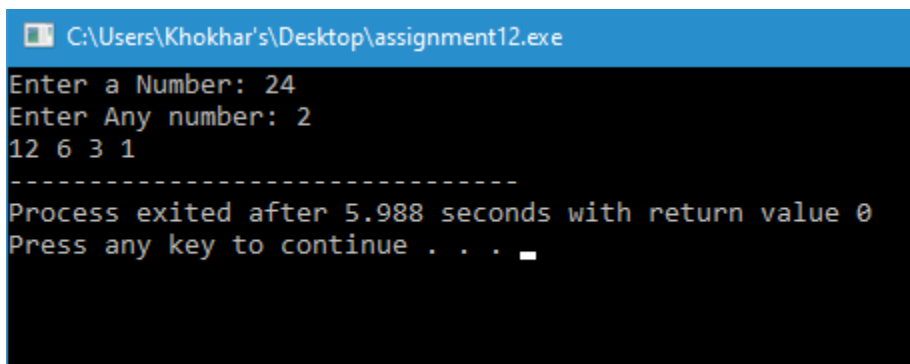
```
C:\Users\Khokhar's\Desktop\assignment12.exe
Enter an octal number : 3452
Decimal equivalent of 3452 is : 1834
-----
Process exited after 11.93 seconds with return value 0
Press any key to continue . . .
```

Program no:4

Write a Program using integers usernum and x as input, then output usernum divided by x four times:



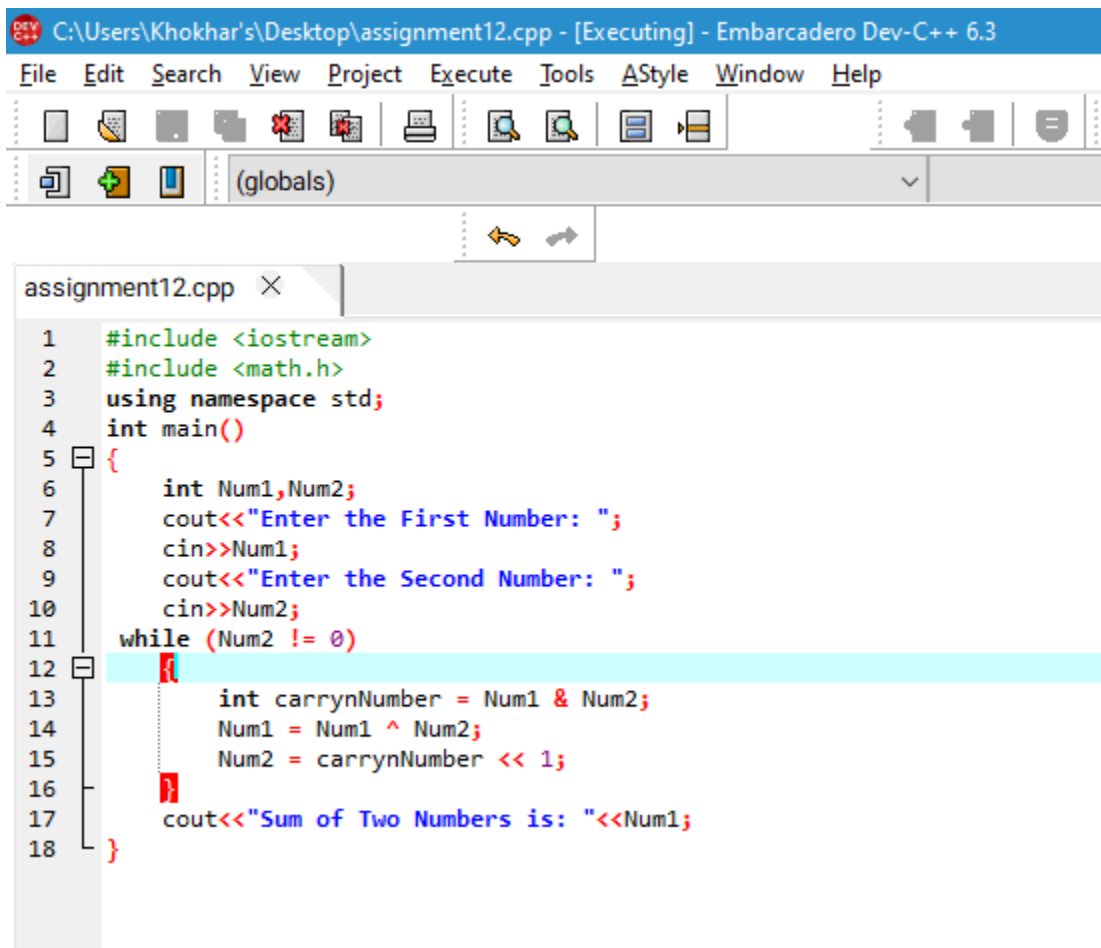
```
1  #include <iostream>
2  #include <math.h>
3  using namespace std;
4  int main()
5  {
6      long num;
7      long num1=0;
8      cout << "Enter a Number: ";
9      cin >> num;
10     while (num1==0)
11     {
12         cout << "Enter Any number: ";
13         cin >> num1;
14     }
15     for (int iloop=0; iloop<4; iloop++)
16     {
17         num /= num1;
18         cout << num << " ";
19     }
20 }
```



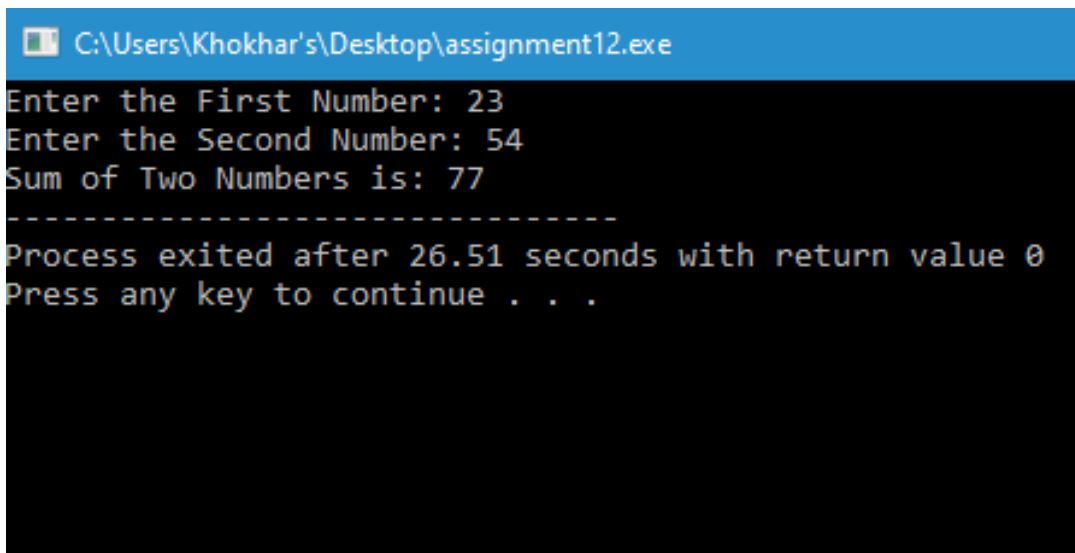
```
C:\Users\Khokhar's\Desktop\assignment12.exe
Enter a Number: 24
Enter Any number: 2
12 6 3 1
-----
Process exited after 5.988 seconds with return value 0
Press any key to continue . . .
```

Program no:5

Write a Program to add two numbers without using addition operator:



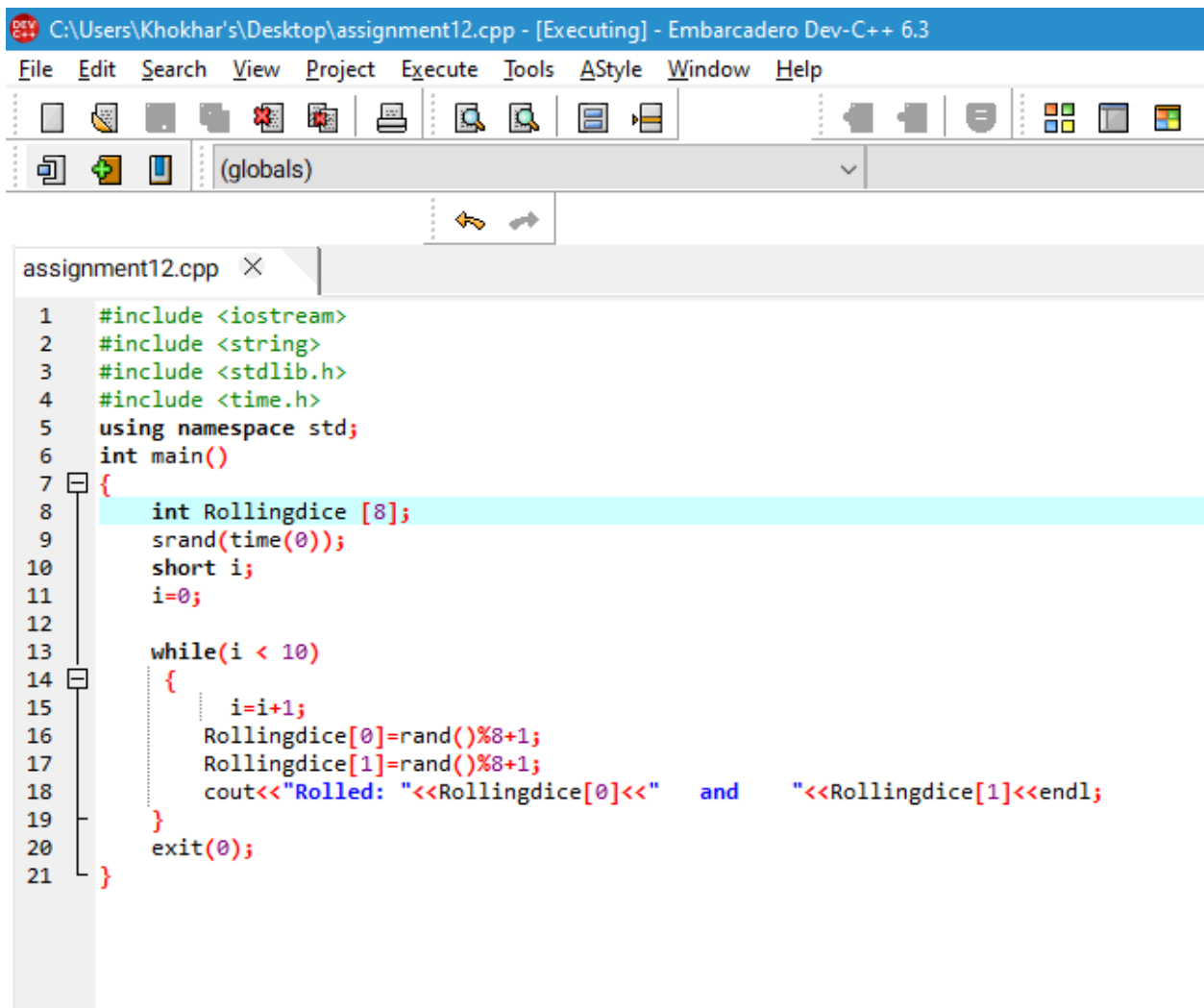
```
C:\Users\Khokhar's\Desktop\assignment12.cpp - [Executing] - Embarcadero Dev-C++ 6.3
File Edit Search View Project Execute Tools AStyle Window Help
(globals)
assignment12.cpp
1  #include <iostream>
2  #include <math.h>
3  using namespace std;
4  int main()
5  {
6      int Num1,Num2;
7      cout<<"Enter the First Number: ";
8      cin>>Num1;
9      cout<<"Enter the Second Number: ";
10     cin>>Num2;
11     while (Num2 != 0)
12     {
13         int carrynNumber = Num1 & Num2;
14         Num1 = Num1 ^ Num2;
15         Num2 = carrynNumber << 1;
16     }
17     cout<<"Sum of Two Numbers is: "<<Num1;
18 }
```



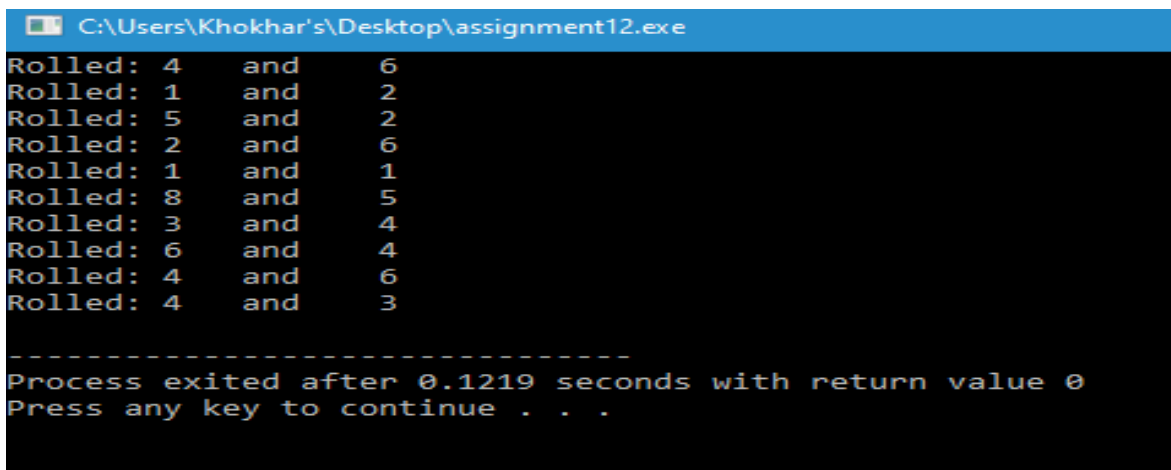
```
C:\Users\Khokhar's\Desktop\assignment12.exe
Enter the First Number: 23
Enter the Second Number: 54
Sum of Two Numbers is: 77
-----
Process exited after 26.51 seconds with return value 0
Press any key to continue . . .
```

Program no:6

Write a Program that simulates the rolling of two dice:



```
1  #include <iostream>
2  #include <string>
3  #include <stdlib.h>
4  #include <time.h>
5  using namespace std;
6  int main()
7  {
8      int Rollingdice [8];
9      srand(time(0));
10     short i;
11     i=0;
12
13     while(i < 10)
14     {
15         i=i+1;
16         Rollingdice[0]=rand()%8+1;
17         Rollingdice[1]=rand()%8+1;
18         cout<<"Rolled: "<<Rollingdice[0]<<"   and   "<<Rollingdice[1]<<endl;
19     }
20     exit(0);
21 }
```



```
C:\Users\Khokhar's\Desktop\assignment12.exe
Rolled: 4      and      6
Rolled: 1      and      2
Rolled: 5      and      2
Rolled: 2      and      6
Rolled: 1      and      1
Rolled: 8      and      5
Rolled: 3      and      4
Rolled: 6      and      4
Rolled: 4      and      6
Rolled: 4      and      3
-----
Process exited after 0.1219 seconds with return value 0
Press any key to continue . . .
```